Name:_____

Exam Question Booklets
OCR Physics A

Booklet 10
14 Questions
83 marks

Things to remember:

✓

✓

✓

✓

✓

✓

✓

1

(a) State what is meant by the diffraction of waves.

.....

.....

.....

.....

..... [2]

(b) Fig. 6.1 shows water ripples approaching a gap in a barrier. In **(i)** the gap is narrow and in **(ii)** the gap is wide.

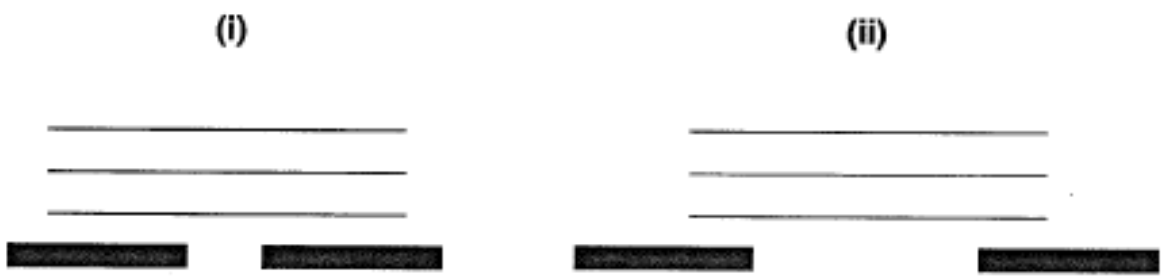


Fig. 6.1

On Fig. 6.1 draw the appearance of the ripples after they have passed through the gap. [3]

(a) **(i)** State **three** phenomena that are associated with **all** waves.

1. .....

2. .....

3. ..... [3]

(ii) State a phenomenon that is only associated with transverse waves.

..... [1]

(b) Define the following wave characteristics.

(i) frequency f

.....

.....

(ii) wavelength λ

.....

.....

[2]

(c) Use the definitions of f and λ to deduce the relationship between f, λ and the speed v of a wave. Explain each stage of your deduction.

[2]

(d) Fig. 3.1 represents the screen of a cathode ray oscilloscope (c.r.o.).

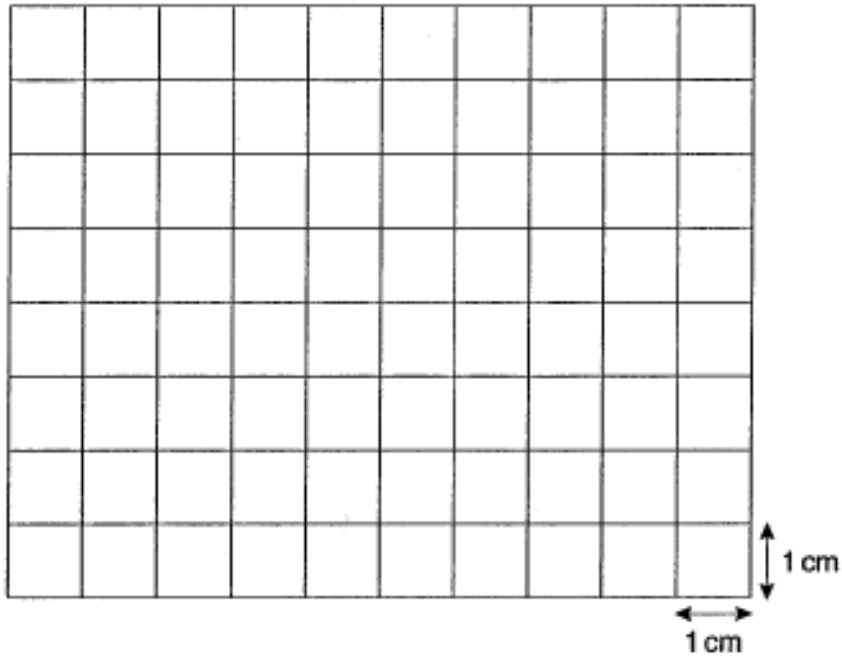


Fig. 3.1

The time-base setting is $0.50\,\text{ms}\,\text{cm}^{-1}$ and the voltage (y-gain) setting is $2.0\,\text{mV}\,\text{cm}^{-1}$. A microphone connected to the c.r.o. detects a pure (sinusoidal) sound wave note of frequency 500 Hz.

(i) Calculate the period of the note.

period = ... s [1]

(ii) The amplitude of the signal from the microphone produced by the note is 6.0 mV.

Draw on Fig. 3.1 the trace produced on the c.r.o. screen when the microphone detects the sound wave. Draw at least two full cycles of the wave on Fig. 3.1. [3]

(iii) The speed of sound in air is $330\,\text{m}\,\text{s}^{-1}$. Calculate the wavelength of the sound received by the microphone.

wavelength = ... m [2]

Fig. 3.1 shows the displacement-time graph for a particle in a medium as a progressive wave passes through the medium.

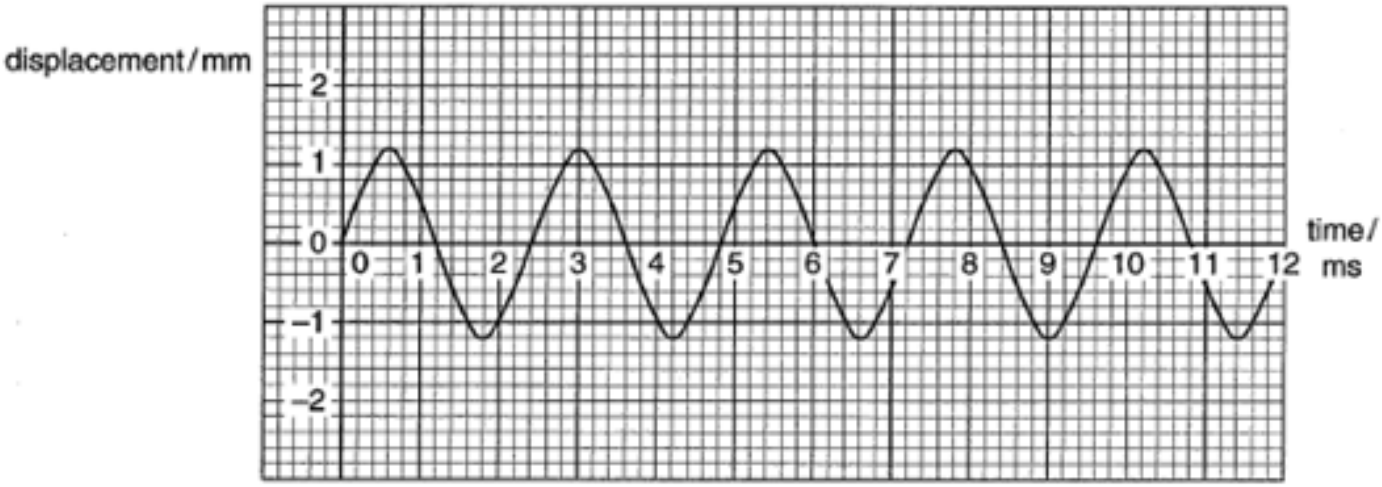


Fig. 3.1

(a) Determine from the graph

(i) the amplitude of the wave

amplitude = ... mm [1]

(ii) the period of the wave.

period = ... ms [1]

(b) **(i)** What is the frequency of the wave?

frequency = ... Hz [2]

(ii) The speed of the wave is $1500\,\mathrm{m\,s^{-1}}$. Calculate its wavelength.

wavelength = ... m [2]

(iii) Use the grid in Fig. 3.2 to sketch a displacement-position graph for the wave at a particular instant. Mark the scale on the position axis and draw at least two full cycles. [3]

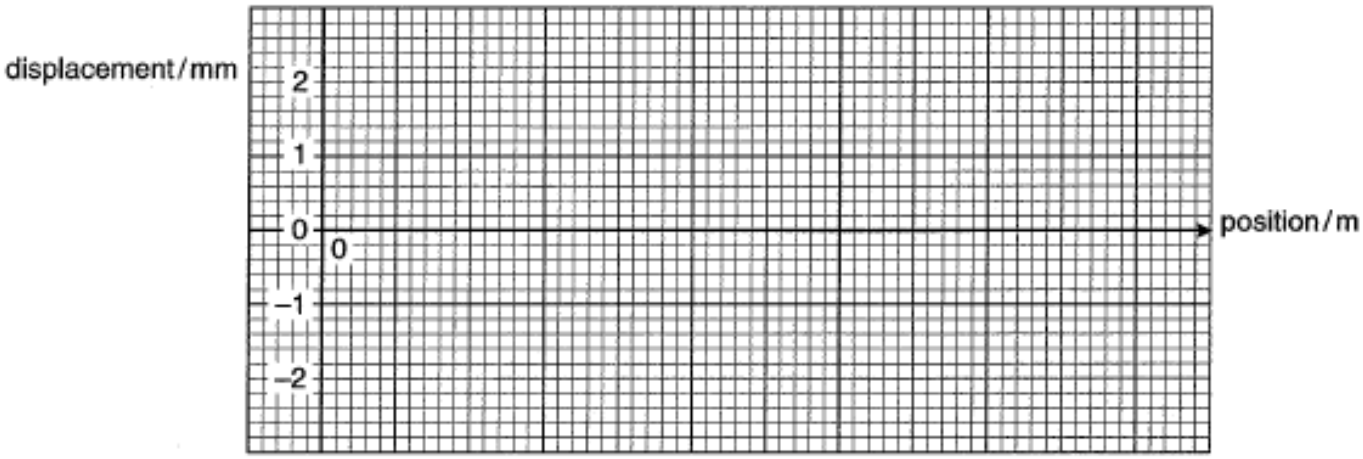


Fig. 3.2

(a) Fig. 3.1 shows a laboratory microwave transmitter T positioned directly opposite a microwave detector D which is connected to a meter.

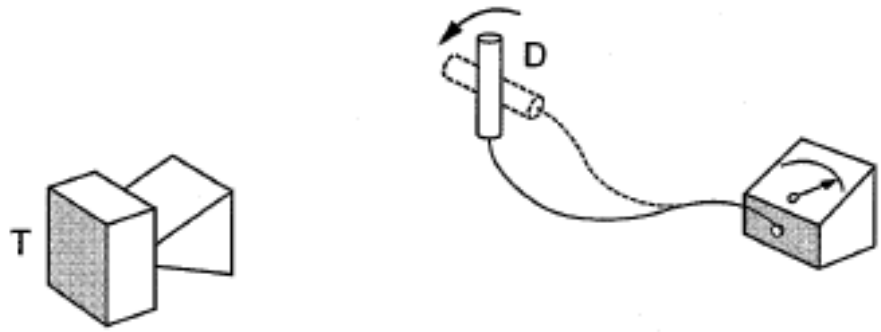


Fig. 3.1

Initially the meter shows a maximum reading. When the detector is rotated through 90°, in a vertical plane as shown, the meter reading falls to zero.

(i) Explain why the meter reading falls.

.....

.....

.....[2]

(ii) Predict what would happen to the meter reading if the detector were rotated through a further 90°.

.....[1]

(iii) State what the observations tell you about the nature of microwaves.

.....[1]

5

Ultra-violet radiation from the Sun is often divided into three regions UV-A, UV-B and UV-C.

(i) Describe the characteristics and dangers of UV-A, UV-B and UV-C radiations.

.....

.....

.....

.....

.....

..... **[3]**

(ii) Explain how sunscreen protects the human skin.

.....

.....

..... **[1]**

A simple cell may be constructed by inserting into a fresh lemon two electrodes made from different metals. The juice of the lemon acts as an electrolyte (conducting liquid). Positive and negative ions within the lemon move towards the metal electrodes. Fig. 2.1 shows such a lemon-cell. It has an e.m.f. of 1.32 V and can provide enough electrical energy to activate a digital clock for many days.

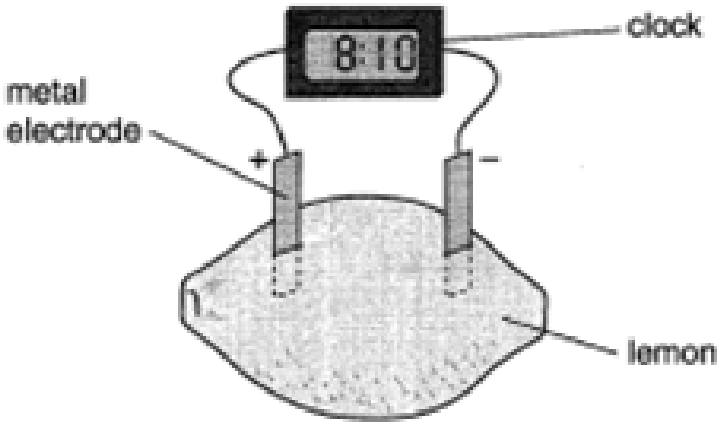


Fig. 2.1

(a) On Fig. 2.1, indicate with an arrow the direction in which negative charge moves within the lemon. [1]

(b) The lemon-cell is capable of providing a steady current of 1.2 mA for eight days (6.9×10^5 s). Calculate

(i) the charge passing through the clock during eight days

charge = ... C [3]

(a) Name an instrument used to measure

(i) electric current,

..... [1]

(ii) potential difference.

..... [1]

(b) The electric charge ΔQ passing a point in a circuit is given by the equation

$$\Delta Q = I\Delta t.$$

State what is represented by the other symbols I and Δt.

I: .....

Δt: ..... [2]

(c) A 1.2 kW water heater is switched on for 1500 s. During this time, a charge of 7.5×10^3 C passes. Calculate

(i) the electric current,

current = ... A [2]

(a) Define electrical resistivity.

.....

..... [2]

(b) Fig. 3.1 illustrates a metallic resistor constructed by depositing a thin layer of metal on a plastic strip. This particular resistor has resistance 5.0 Ω, length 1.2×10^{-2} m and width 2.0×10^{-3} m.

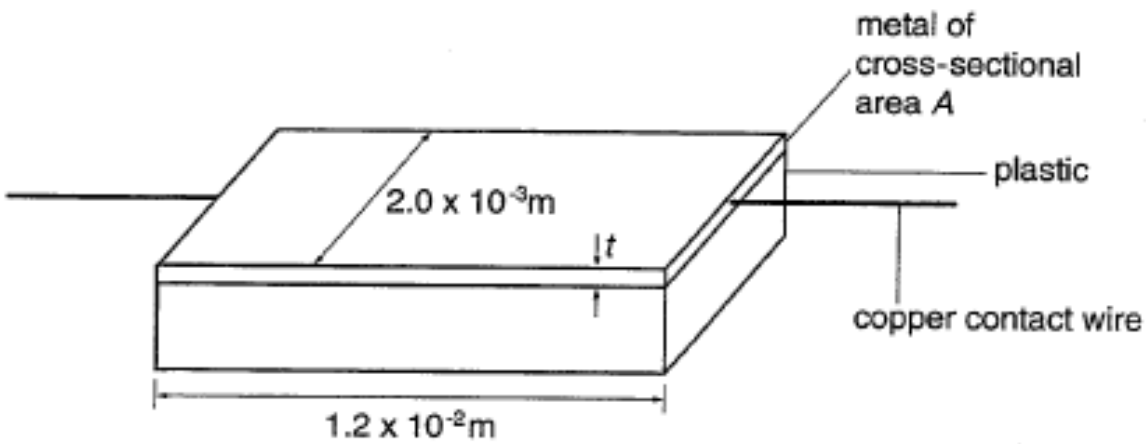


Fig. 3.1

(i) The resistivity of the metal is 4.3×10^{-6} Ω m. Calculate the cross-sectional area A of the resistor.

A = ... m^2 [3]

(ii) What is the thickness t of the resistor?

t = ... m [1]

9

(a) **(i)** Use energy considerations to distinguish between potential difference (p.d.) and electromotive force (e.m.f.)

p.d.: .....

.....

.....

e.m.f.: .....

.....

..... [2]

(ii) Place a tick (✓) in the appropriate box opposite the correct answer for an alternative unit for e.m.f. or p.d.

$J\,s^{-1}$ ☐

$J\,A^{-1}$ ☐

$J\,C^{-1}$ ☐ [1]

(a) Define electrical resistance.

..... [2]

(b) **(i)** State Ohm's law.

.....

.....

..... [2]

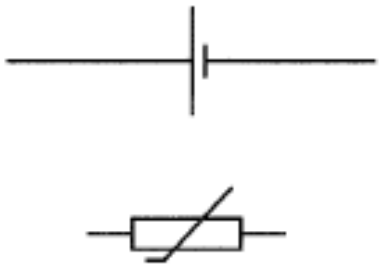
(ii) Place a tick (✓) in the box for any component that obeys Ohm's law and place a cross (✗) for any that does not.

metallic resistor ☐

diode ☐

thermistor ☐ [2]

(c) A cell is connected across a thermistor. Using appropriate circuit symbols, complete Fig. 2.1 to show how the current in the thermistor and the p.d. across it may be measured.



[2]

11

Define *potential difference*.

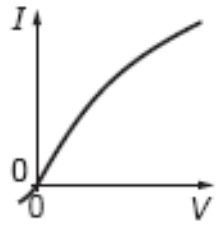
.....

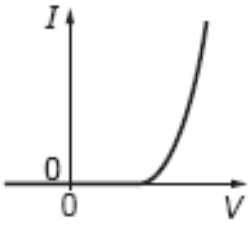
.....[1]

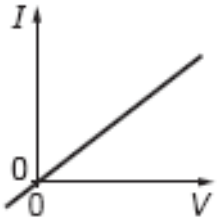
12

(a) Draw a line from each of the named components on the left-hand side to the correct *I-V* characteristic on the right-hand side.

metallic wire at constant temperature
semiconductor diode
filament lamp







[1]

(b) In this question, two marks are available for the quality of written communication.

Explain, in terms of resistance, the shape of the *I-V* graph for each component below.

(i) Metallic wire at constant temperature.

.....

.....

.....

.....

.....

(ii) Semiconductor diode.

.....

.....

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.....

.....

(iii) Filament lamp.

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.....

.....

[6]

Fig. 3.1 shows the I-V characteristic of a particular electrical component.

13

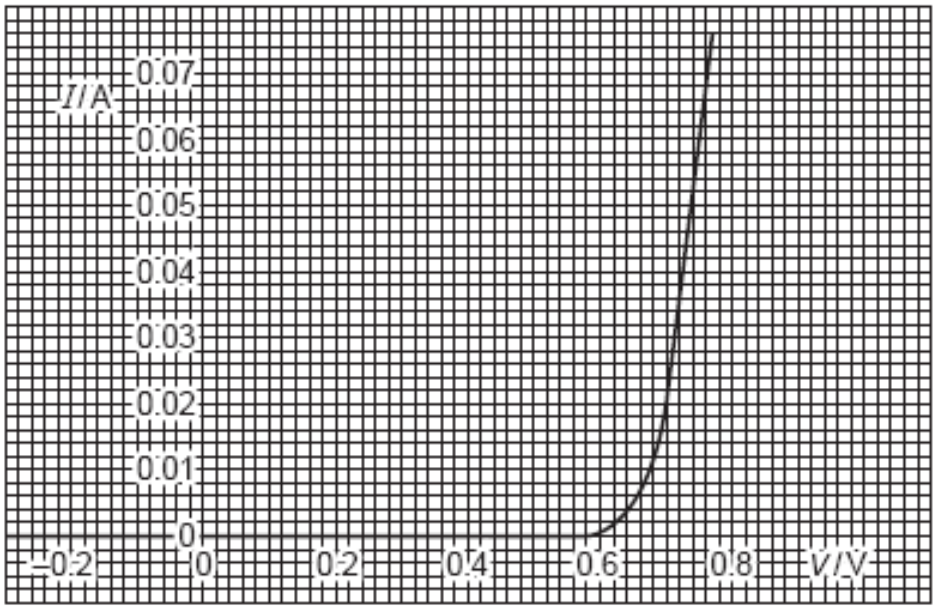
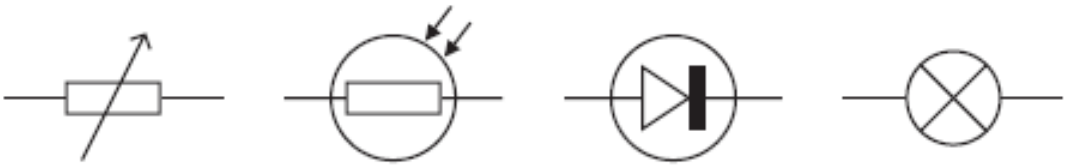


Fig. 3.1

(a) Name the component.

..... [1]

(b) Circle the correct circuit symbol for the component. [1]



(c) Use Fig. 3.1 to calculate the resistance of the component at 0.20V and 0.70V.

resistance at 0.20V = ... Ω

resistance at 0.70V = ... Ω [3]

(a) Define electrical *resistivity*.

.....

..... [2]

(b) Fig. 4.1 shows a conducting paint in a cylindrical glass vessel.

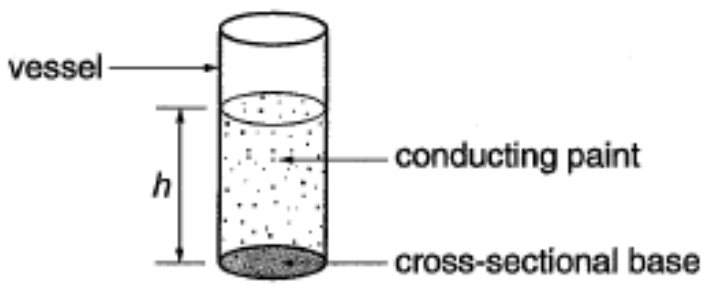


Fig. 4.1

The volume of the paint is $1.2 \times 10^{-5}\,m^3$ and the vessel has base of area $3.0 \times 10^{-4}\,m^2$.

(i) Show that the height h of the paint column is 4.0 cm.

[1]

(ii) Calculate the resistance of the paint column of height 4.0 cm. The resistivity of the paint is $6.9 \times 10^{-2}\,\Omega\,\text{m}$.

resistance = ... Ω [2]

(c) State and explain how your answer to **(b)(ii)** changes when the same volume of paint is poured into a cylindrical glass vessel having a base of double the cross-sectional area.

.....

.....

.....

.....

..... [2]